



QUESTIONS

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Fib series-iterative & recur



FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

Iterative: 0 1 1 2 3 5 8

Recursive: 0 1 1 2 3 5 8

Recursive Calls: 13

Iterative Time: 0.001 ms

Recursive Time: 0.005 ms

Test Case:

Enter N: 1

Output:

0

PROGRAM EDITOR

C



Run

Save

```
1 #include <stdio.h>
2 #include <time.h>
3 int calls = 0;
4 int fib(int n) {
5     calls++;
6     if (n <= 1)
7         return n;
8     return fib(n - 1) + fib(n - 2);
9 }
10 int main() {
11     int n, i;
12     int a = 0, b = 1, c;
13     clock_t start, end;
14     printf("Enter N: ");
15     scanf("%d", &n);
16     start = clock();
17     printf("Iterative: ");
18     for (i = 0; i < n; i++) {
19         printf("%d ", a);
20         c = a + b;
21         a = b;
```

OUTPUT

```
Enter N: Iterative: 0 1 1 2 3 5 8
Iterative Time: 0.000 ms
Recursive: 0 1 1 2 3 5 8
Recursive Calls: 59
Recursive Time: 0.000 ms
```